Quadratics: Expanding & Factorising

Practice test 4 · 40 marks · Higher tier · non-calculator

Name: _____ Date: _____ Score: _____ / 40

Section A — Expanding double brackets

A1. Expand and simplify $(x – 9)(x – 4)$

[2 marks]

A2. Expand and simplify $(5 – x)(3 + x)$

[2 marks]

A3. Expand and simplify $(3x – 2y)^2$

[2 marks]

A4. Expand and simplify $(2x + 1)(x – 3) – (x – 2)^2$

[2 marks]

Section B — Expanding three brackets

B1. Expand and simplify $(x + 5)(x – 5)(x + 2)$

[3 marks]

B2. Expand and simplify $(3x - 1)(x + 2)^2$

[3 marks]

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Factorise $x^2 - 16x + 63$

[2 marks]

C2. Factorise $x^2 + 5x - 84$

[2 marks]

C3. Factorise $4x^2 - 1$

[1 mark]

C4. Factorise fully $x^4 - 81$

[2 marks]

Section D — Factorising $ax^2 + bx + c$

D1. Factorise $6x^2 + x - 2$

[2 marks]

D2. Factorise $10x^2 - 17x + 3$

[2 marks]

D3. Factorise fully $8x^2 - 2x - 6$

[2 marks]

Section E — Using factorising

E1. Solve $3x^2 - 5x = 2$

[3 marks]

E2. Simplify fully

$$\frac{4x^2 - 9}{2x^2 + x - 6}$$

[3 marks]

Section F — Problem solving & proof

F1. Write $2x^2 + 12x + 7$ in the form $a(x + b)^2 + c$

[3 marks]

F2. n is an integer. Prove that $(2n + 1)^2 - (2n - 1)^2$ is always a multiple of 8.

[3 marks]

F3. Without a calculator, work out $8.7^2 - 1.3^2$

[1 mark]